The stable positional fifty-three factor and its coherent divisors

E.123. Actual positional class norms, unit defects and a complete coherent-divisor image [Proved]

Keep the compatible roots of E.119–E.122, and put

$$\begin{gathered} K=\mathbb{Q}(\zeta_3),\quad \xi=\zeta_9,\quad \beta_d^{3^d}=53\zeta_3,\quad \beta_{d+1}^3=\beta_d,\\ L_d=\mathbb{Q}(\beta_d),\quad B_d=L_d(\xi),\quad C_d=\mathrm{Cl}_S(L_d)\{3\},\quad V_d=\mathcal{O}_{L_d,S}^\times,\qquad S\mid\{3,53\}. \end{gathered} \tag{E123.1}$$

For every $d\ge 2$, the horizontal extension B_d/L_d is cyclic, unramified everywhere and split at all places in S. There is one place of L_d over each of 3 and 53, with residue degrees one and two respectively. Both are totally ramified in L_{d+1}/L_d. The actual vertical ideal norms, extensions and automorphisms satisfy

$$\begin{gathered} N_d^L: C_{d+1}\xrightarrow{\sim} C_d,\qquad C_d\simeq C_9^2,\qquad \sigma_d|_{C_{d+1}}=1,\\ m_2=1,\quad m_d=N_2^L\cdots N_{d-1}^L,\qquad m_dN_d^L=m_{d+1},\quad m_{d+1}j_d^L=3m_d,\\ \varprojlim_{d,N_d^L} C_d\xrightarrow[\mathrm{pr}_2]{\sim} C_2,\qquad j_d^L=3(N_d^L)^{-1}. \end{gathered} \tag{E123.2}$$

The inverse in the last formula is the inverse of the displayed isomorphism. In the actual E.121 bases on $A_2=C_9\oplus C_3^3$ and $C_2=C_9u\oplus C_9a$, every horizontal norm and extension, transported by n_d and m_d, is

$$N_h=\begin{pmatrix}1&0&6&6\\0&3&6&3\end{pmatrix},\qquad j_h=\begin{pmatrix}0&3\\2&0\\2&0\\0&1\end{pmatrix}. \tag{E123.3}$$

The Artin map for B_d/L_d, in these coordinates, is $(r,s)\mapsto s \bmod 3$, with the positive generator $h(\xi)=\xi^4$.

For every $d\ge 2$, all later one-step unit norm quotients are evaluated:

$$\begin{gathered} V_d/N_{L_{d+1}/L_d}V_{d+1}\simeq C_3^3,\qquad \frac{V_d\cap N_{L_{d+1}/L_d}L_{d+1}^\times}{N_{L_{d+1}/L_d}V_{d+1}}\simeq C_3^2,\\ V_d/N_{B_d/L_d}U_d\simeq C_3,\qquad U_d=\mathcal{O}_{B_d,S}^\times. \end{gathered} \tag{E123.4}$$

Their forward maps and inverses are the actual Hilbert–90 divisor maps in the proof. The kernels after completion for multiplication by 3^n are, respectively,

$$\ker(\widehat{V}_{d+1,3}\to\widehat{V}_{d,3})\simeq\mu_3\oplus\mathbb{Z}_3^{2\cdot 3^d},\qquad \ker(\widehat{U}_{d,3}\to\widehat{V}_{d,3})\simeq\mu_3\oplus\mathbb{Z}_3^{2\cdot 3^d+4}. \tag{E123.5}$$

The torsion factors are literal roots of unity. The proof specifies the chosen basis maps, inverse maps and their ambiguity.

Here is the entire coherent divisor image for this positional factor. Define

$$\begin{gathered} \mathcal{P}=\{p\ne 3,53 : v_3(\mathrm{ord}_{\mathbb{F}_p^\times}(53))=1\},\\ \mathcal{R}(\mathcal{P})=\{(v_p)\in\textstyle\prod_{p\in\mathcal{P}}\mathbb{Z}_3 : v_p\equiv 0 \pmod{3^n}\text{ for almost all }p,\text{ for every }n\}. \end{gathered} \tag{E123.6}$$

Each $p\in\mathcal{P}$ has exactly one branch $(\mathfrak{p}_d)_{d\ge 2}$ of residue degree one that survives the completed ideal-norm limit. Put $c_p=m_d[\mathfrak{p}_d]\in C_2$. The class is independent of d. The sum below is finite modulo nine:

$$c:\mathcal{R}(\mathcal{P})\longrightarrow C_2,\qquad c(v)=\sum_p c_pv_p,\qquad \mathcal{D}=\ker c. \tag{E123.7}$$

The map c is onto. Concrete pivot primes are 31 and 103. At level two their residues are

p	$\beta_2 \bmod \mathfrak{p}_2$	$\beta_1 \bmod \mathfrak{p}_1$	$p_L(c_p) \in C_9 \oplus C_3$	$c_p \bmod 3$ in (u,a)
31	27	29	$(4,1)$	$(0,1)$
103	80	90	$(4,0)$	$(1,1)$

$$p_L = \begin{pmatrix} 6 & 1 \\ 2 & 1 \end{pmatrix}. \tag{E123.8}$$

Thus (c_{31}, c_{103}) itself is an actual basis of C_2. The literal HNF ideals, seven full principal ideal equalities, complete relation lattice and inverse coordinate maps are retained in the certificate; the certification uses no upper-field BNF class-group assumption.

Write $c_p = a_p c_{31} + b_p c_{103}$, choosing $a_p, b_p \in \{0, \dots, 8\}$, and put $D_p = e_p - a_p e_{31} - b_p e_{103}$ for $p \neq 31, 103$. Then the following maps are inverse continuous homomorphisms:

$$\begin{aligned} \mathbb{Z}_3^2 \times \mathcal{R}(\mathcal{P} \setminus \{31,103\}) &\longrightarrow \mathcal{D}, \\ (A,B,t) &\longmapsto 9Ae_{31} + 9Be_{103} + \sum_{p \neq 31,103} t_p D_p, \\ v &\longmapsto \left(\frac{v_{31} + \sum_p a_p v_p}{9}, \frac{v_{103} + \sum_p b_p v_p}{9}, (v_p)_{p \neq 31,103} \right). \end{aligned} \tag{E123.9}$$

Both sums in the inverse omit $31, 103$. All coefficients are bounded, so these sums converge in $\mathbb{Z}_3$.

Put $\mathcal{X} = \varprojlim_{d,N} \widehat{L_{d\,3}^{\times}}$ and $\mathcal{V} = \varprojlim_{d,N} \widehat{V}_{d,3}$. There is an exact sequence of topological groups

$$0 \longrightarrow \mathcal{V} \longrightarrow \mathcal{X} \xrightarrow{\operatorname{div}_{S^c}} \mathcal{D} \longrightarrow 0. \tag{E123.10}$$

For every basis vector in (E123.9), choose an actual coherent lift A_{31}, A_{103}, A_p with the indicated divisor. Then

$$\begin{aligned} s(v) &= A_{31}^{A} A_{103}^{B} \prod_{p \neq 31,103} A_p^{t_p}, \\ \mathcal{V} \times \mathcal{D} &\xrightarrow{\sim} \mathcal{X}, \qquad (y,v) \longmapsto y\, s(v), \\ x &\longmapsto (x / s(\operatorname{div}_{S^c} x), \operatorname{div}_{S^c} x) \end{aligned} \tag{E123.11}$$

are actual maps and inverse maps. Every infinite product converges in each field-level power-completion topology. The compact group $\mathcal{V}$ is torsion-free and is isomorphic to $\prod_{i \geq 1} \mathbb{Z}_3$ by the explicit character construction in the proof. Its actual norm to $\widehat{\mathbb{Q}^{\times}}_3$ is retained, rather than replaced by an abstract product coordinate.

More precisely, let $\mathfrak{q}_3, \mathfrak{q}_{53}$ be the two S-primes of L_2, and define continuous homomorphisms

$$\begin{aligned} \nu_3(y) &= v_{\mathfrak{q}_3}(y_2), \qquad \nu_{53}(y) = 2v_{\mathfrak{q}_{53}}(y_2), \\ \rho(v) &= v_{\mathfrak{q}_3}(s(v)_2), \qquad \eta(v) = 2v_{\mathfrak{q}_{53}}(s(v)_2), \\ N_{\mathbb{Q}}(y\, s(v)) &= 3^{\,\nu_3(y)+\rho(v)} 53^{\,\nu_{53}(y)+\eta(v)} \prod_{p \in \mathcal{P}} p^{v_p}. \end{aligned} \tag{E123.12}$$

The family $(\beta_d)_d$ is an actual element of $\mathcal{V}$ with $(\nu_3, \nu_{53}) = (0,2)$. Equation (E123.12), together with the unchanged norms of the other positional factors, is the literal total-norm map in E.46. In particular $\mathcal{D}$ embeds as a closed family of actual H-section classes on every first-coordinate line $3^k \kappa_{53}$ containing this factor: set this factor to $s(v)^3$ and all other factors to one. Its scaled second toric coordinate is

$$\rho(v)\kappa_3 + \eta(v)\kappa_{53} + \sum_{p \in \mathcal{P}} v_p \kappa_p. \tag{E123.13}$$

The inverse on this family is the outside-S divisor divided by three. The proof keeps the factor map, the division by three and the original total-norm cube condition explicit. The other positional factors and the further intrinsic quotient maps in E.23 retain their previous exact definitions.

Proof. The fields and actual class-field maps. The degree of L_d/K is 3^d, by the valuation-one Eisenstein equation at the prime over 53. The same argument over $\mathbb{Q}(\xi)$ gives $[B_d : L_d] = 3$. For $d \geq 2$, $B_d = B_2 L_d$, so E.121's actual unramified, S-split extension B_2/L_2 base-changes to the stated extension B_d/L_d. The automorphism h cycles the three primes in each of E.122's two triples. Its quotient therefore has one prime in each case; the residue degrees and vertical total ramification follow from the literal norm and restriction maps on these primes. This also proves the asserted absence of other ramification.

Let H_D^S denote the maximal unramified abelian three-extension of D in which every S-prime splits. The actual inclusions are

$$B_d \subset H_{L_d}^S \subset H_{B_d}^S, \qquad H_{L_d}^S = \text{the maximal subfield of } H_{B_d}^S \text{ abelian over } L_d. \tag{E123.14}$$

Indeed unramified and S-split extensions remain so under base change. Conversely every abelian intermediate field over L_d in the right side is unramified and S-split over L_d, because B_d/L_d has those properties. For an extension of a group A by a cyclic group generated by h, with A abelian, the commutator subgroup is $(h-1)A$: choose one lift of h and commute it with A; commutators of its powers give no additional subgroup. Artin reciprocity and (E123.14) thus give the exact sequence

$$0 \longrightarrow (A_d)_h \xrightarrow{N_h} C_d \xrightarrow{\mathrm{Art}_{B_d/L_d}} C_3 \longrightarrow 0. \tag{E123.15}$$

The arrow on the left is the actual ideal norm. Its agreement with restriction of Frobenius is Milne V Proposition3.3, including its proof. The class-field identification uses V Theorem3.6 and Example3.9 [1].

E.122 transports the entire h-action to the matrix H of E.121. Its coinvariants have order 27, so (E123.15) gives $|C_d| = 81$ for all $d \geq 2$. Moreover $H_{L_d}^S \cap L_{d+1} = L_d$: any nontrivial intersection would be the cyclic cubic L_{d+1}/L_d, which is ramified at both S-places. For completeness, the actual norm surjectivity can be read directly from this intersection. For a prescribed Frobenius in $H_{L_d}^S/L_d$, choose its extension acting identically on L_{d+1} in the compositum, then choose a prime with that Frobenius. A prime above it has relative residue degree one, so its ideal norm has the prescribed class. This is the exact norm/Frobenius square of Milne V3.3 and the Chebotarev theorem VIII7.4. Equal orders now make every N_d^L an isomorphism. The identity $N_d^L \sigma_d = N_d^L$ proves that σ_d acts identically; norm followed by extension is the trace $1 + \sigma_d + \sigma_d^2 = 3$. This proves (E123.2).

Norms commute in the field tower. Norm and extension in the horizontal-vertical square commute because $B_d \cap L_{d+1} = L_d$, the first extension being S-split and the second totally ramified there, and their compositum is B_{d+1}. On an ideal this equality is the equality of the residue-degree weighted prime sums. Thus E.121's four complete class maps transport to (E123.3). In particular their Artin, kernel and image maps are retained; no inverse of a noninjective norm is used.

There is also a useful literal local comparison. Write $53 = -(1 - 54)$ and define $a_1 = -\exp(\log(1-54)/9) \in \mathbb{Q}_3$. The logarithm has valuation three, so $a_1^9 = 53$ and $v_3(a_1 + 1) = 1$. At every labeled completion of B_2 over three, $(\beta_2/a_1)^9 = \zeta_3$, hence β_2/a_1 is a primitive twenty-seventh root of unity. The local degrees give

$$(B_2)_v = (L_2)_w = \mathbb{Q}_3(\mu_{27}). \tag{E123.16}$$

This is a map of the actual completed fields. Globally $\mu(L_d)\{3\} = \mu_3$: otherwise $\xi \in L_d$, contradicting $[B_d : L_d] = 3$. E.122 gives $\mu(B_d)\{3\} = \mu_9$.

An actual unit character, and every later unit quotient. Use the literal root v_{20} of E.122 in $\mathbb{Q}[b,z]/(b^9 - 53z^3, z^6 + z^3 + 1)$, and put

$$w = v_{20}(b,z)v_{20}(b,z^4)v_{20}(b,z^7) \in L_2^\times. \tag{E123.17}$$

The rational-ring certificate rechecks its full cube identity, each horizontal substitution, and its complete norm. The output gives every coefficient of the resulting polynomial of degree at most seventeen in $b = \beta_2$. A separate no-input GP program checks that polynomial by exact resultants. The decisive values are

$$\begin{gathered} N_{L_2/K}(w) = \frac{3+6\zeta_3}{53}, \qquad N_{L_2/\mathbb{Q}}(w) = \frac{27}{53^2}, \\ v_{\mathfrak{q}_{53}}(w) = -1, \qquad \overline{w\beta_2} = 10 + 50\zeta_3 \in \mathbb{F}_{53^2}, \qquad (10+50\zeta_3)^{936} = \zeta_3^2. \end{gathered} \tag{E123.18}$$

Its coefficient denominator divides 53^2. Since its absolute norm has support only at $3, 53$, the whole principal divisor is S-supported: away from S the element is integral, and the positive norm valuations there sum to zero. Thus $w \in V_2$. The positive local Artin convention of E.122 is minus the displayed tame residue logarithm, so $\chi_{53}(w) = 1$. The product formula gives $\chi_3(w) = 2$. This also agrees with the two triple sums of the literal root's six characters.

These two characters remain $(2,1)$ on the included $w \in V_d$ for every $d \geq 2$. At either unique local branch, $N_{(L_d)_v/(L_2)_w}\beta_d = \beta_2$. The cup-product projection formula therefore reads

$$\operatorname{Cor}\big([\beta_d] \cup \operatorname{Res}[w]\big) = [\beta_2] \cup [w], \qquad \operatorname{inv}_{(L_2)_w} \operatorname{Cor} = \operatorname{inv}_{(L_d)_v}. \tag{E123.19}$$

This is the actual restriction/corestriction map with common coefficient μ_3, not multiplication of a character by the extension degree. Milne III3.1 proves the required compatibility by cup products; V5.2 gives the product of local reciprocity maps. The local norm quotient has order three, so the ordinary field-norm obstruction on V_d is exactly the one-dimensional character χ_{53}, with $\chi_3 = -\chi_{53}$. The cyclic Hasse norm theorem VIII3.1, with its injectivity proof, identifies its kernel with $V_d \cap NL_{d+1}^\times$ [1].

Here are the precise unit-defect maps, including the completion step. For a cyclic cubic E/D in this tower, write τ for the positive generator and V_E, V_D for its S-units. Given $u = Nx \in V_D$, the norm of $\operatorname{div}_{S^c}(x)$ is zero. The cyclic permutation-module calculation on every outside-S prime orbit gives an outside-S divisor D_x with

$$\operatorname{div}_{S^c}(x) = (\tau - 1)D_x, \qquad \nu(u) = \operatorname{pr}_3[D_x] \bmod j\operatorname{Cl}_S(D)\{3\}. \tag{E123.20}$$

Changing x with its norm fixed changes it by a Hilbert–90 coboundary, hence changes D_x by a principal divisor. Changing D_x with $(\tau-1)D_x$ fixed adds an invariant divisor. Outside S the extension is unramified, so invariant divisors are precisely extended base divisors. These are the exact ambiguities in (E123.20). Its image is $\operatorname{Cl}_S(E)\{3\}^\tau / j\operatorname{Cl}_S(D)\{3\}$: for an invariant class choose a representative D, choose x with $(\tau-1)D = \operatorname{div}_{S^c}(x)$, and send it back to Nx. Its kernel consists exactly of norms of S-units, by dividing x by the corresponding principal Hilbert–90 coboundary. The prime-to-three part of the invariant quotient vanishes because norm followed by extension is the trace, which is multiplication by three on invariant classes. This proves both directions of the map, also for the horizontal extension.

For the vertical extension, (E123.2) makes this invariant quotient $C_2/3C_2 = C_3^2$. For the horizontal extension all S-units are field norms: at S the extension is split, and elsewhere a unit is a local norm in an unramified extension. In the E.121 coordinates its invariant quotient is

$$A_d^h/j_hC_d \xrightarrow{\sim} \mathbb{F}_3, \qquad (3r,t,t,s) \longmapsto r - s. \tag{E123.21}$$

An inverse is the class of $(3,0,0,0)$ followed by the literal ideal construction in (E123.20). To give coordinates on the full vertical quotient, first subtract the unique power w^e, $e=\chi_{53}(v)\in\{0,1,2\}$. Apply (E123.20) to vw^{-e} and use either actual basis of $C_2/3C_2$. Conversely choose divisors representing that basis, choose the stated principal corrections x_1,x_2, and map (e,r,s) to the class of $w^e(Nx_1)^r(Nx_2)^s$. All carries of three vanish because every included cube is an actual unit norm. This proves (E123.4) and the inverse maps.

The groups of S-units are finitely generated. Completion for multiplication by 3^n preserves their exact norm-image sequence: diagonalize the integer map between the free parts, retain the finite root groups, and complete each displayed cyclic factor. The free ranks are 3^d+1 for V_d and $3^{d+1}+5$ for U_d. The vertical norm on μ_3 is cubing; the horizontal norm on μ_9 sends ξ to $\xi^{1+4+7}=\zeta_3$. This gives precisely (E123.5). For a completed kernel K, its torsion is the stated literal μ_3; choose a $\mathbb{Z}_3$-basis of K/μ_3 and lifts $k_i\in K$. The isomorphism is $(z,(a_i))\mapsto z\prod_i k_i^{a_i}$. The inverse takes quotient coordinates and divides by their lifted product. A change of basis acts by its actual invertible matrix; changing lifts multiplies the first coordinate by the corresponding product of roots of unity.

The two literal prime classes and an unconditional presentation. The lower field L_1 and its complete ideal and unit data are certified in E.119. Its outside-S ideal basis G_1,G_2 and local Artin map give the following direct recipe. For a degree-one prime $\mathfrak{p}_1$, compute the exact equality $\mathfrak{p}_1=G_1^{e_1}G_2^{e_2}(x)$, including all S-valuations. Its actual idelic coordinate is $(-\chi_3(x),-\chi_{53}(x),e_1,e_2)$: the sign follows by applying global reciprocity to (x). Project this vector through E.119's exact kernel lattice and Smith map. For 31 and 103 the calculation gives

p	(e_1,e_2)	$(\chi_3(x),\chi_{53}(x))$	$(-\chi_3(x),-\chi_{53}(x),e_1,e_2)$
31	$(5,0)$	$(1,1)$	$(-1,-1,5,0)$
103	$(2,2)$	$(1,0)$	$(-1,0,2,2)$.

(E123.22)

Every rational coefficient of x, the whole lower HNF prime ideal, and the full principal ideal equality are in the replayable GP output. Applying the exact Smith map gives (E123.8). Reduction of p_L modulo three is invertible; hence those two full classes are independent modulo three and form a basis of the proved group C_9^2.

The residue condition identifies the actual surviving primes, not just arbitrary primes of the same rational norm. Both rational primes are 4 modulo nine and 53 is noncubic. For $m=(p-1)/3$, the unique prime-to-three root is $b_p=53^{3(27^{-1}\bmod m)}\bmod p$. It satisfies $b_p^m=1$, $b_p^{27}=53^3$, and $\zeta_3=b_p^9/53$. Thus $\beta_2=b_p$ specifies the desired degree-one prime; its lower residue is b_p^3. The output verifies these identities for the displayed numbers.

Here is a full literal certification of the upper presentation. The proposed ordinary generators I_1,I_2,I_3 supply seven checked ideal equalities: their powers $9,9,3$, the two S-prime relations, and the two pivot-prime corrections. No correctness assumption about the proposed BNF group is used. Those equalities give an actual homomorphism from

$$\mathbb{Z}^3/\begin{pmatrix}9&0&0&6&0\\0&9&0&3&0\\0&0&3&1&0\end{pmatrix}\mathbb{Z}^5\longrightarrow \mathrm{Cl}_S(L_2). \tag{E123.23}$$

The source has order 81. Both independently proved generating classes c_{31},c_{103} lie in its image by the checked pivot equalities, so the map is an isomorphism onto the entire three-primary subgroup. This proves the required certification without the interrupted upper-field certification attempt. The exact maps between generator coordinates and the prime basis are

$$P=\begin{pmatrix}7&7&0\\2&4&3\end{pmatrix},\qquad Q=\begin{pmatrix}5&4\\8&5\\1&0\end{pmatrix},\qquad PQ=I_2\pmod 9. \tag{E123.24}$$

The inverse composition differs from the identity by the relation lattice in (E123.23). The full Smith matrices and this equality are checked over the integers. All HNF matrices are in the literal integral basis $1, \zeta_3, \beta_2, \zeta_3\beta_2, \ldots, \beta_2^8, \zeta_3\beta_2^8$. It is maximal: the valuation-one Eisenstein equation at 53 is tame of degree nine over K; (E123.16) proves maximality at three, and the polynomial discriminant has no other prime support. The discriminant of the stated basis is $-3^{45}53^{16}$, as checked independently. The candidate object is used only to propose exact ideal witnesses, each of which is verified. A further standalone program reads no file and performs no BNF operation: it reconstructs the two number fields from the literal polynomials and bases, obtains empty `nfcertify` obstruction lists, and rechecks all nine lower and upper principal ideal equalities, both positive Artin columns, the chosen prime residues, and the full forward and inverse relation maps.

For any other prime in $\mathcal{P}$, its coordinate c_p can therefore be evaluated by an exact ideal relation in this certified finite presentation. One unconditional terminating procedure enumerates the 81 prime-basis classes, integral powers of the two S-primes, and nonzero field elements, until the full ideal equality is found; its class is then unique. The retained principal-ideal routine proposes this equality directly and accepts it only after full ideal verification. This specifies the actual finite class map rather than assigning the missing kernel coordinate from the genus projection.

Surviving branches, completion and coherent lifts. At a prime away from $3, 53$, Kummer roots identify prime branches with Frobenius orbits on the compatible 3-power roots of $53\zeta_3$. On a fixed cyclotomic component the action is affine on $\mathbb{Z}_3$. Its only bounded orbits are fixed points: if a positive power fixes x, then $(q^n - 1)(x - x_0) = 0$ in the torsion-free ring $\mathbb{Z}_3$. The required divisibility is uniform on each finite node away from the fixed point. After a finite refinement such a node is a residue ball on which $v_3(x - x_0) = t$ is constant. Replace the residue Frobenius by its square when necessary, so $q \equiv 1 \pmod 3$, and put $s = v_3(q - 1)$. The exact formula $\phi^r(x) - x = (q^r - 1)(x - x_0)$ and the elementary lifting-the-exponent identity give orbit three-part $3^{\max(0,m-s-t)}$ modulo 3^m, uniformly on that ball. If $x_0 \notin \mathbb{Z}_3$, the valuation is constant on every integral ball after the same refinement. Thus the norm from sufficiently deep descendants of any fixed nonfixed node has an arbitrarily large common three-power weight. Its coherent integral mass is divisible by every power of three and is zero. In the nested nodes containing x_0, all other descendants vanish by this argument; precisely the fixed descendant retains its mass. For a split rational prime the root of unity component of 53 is canceled by exactly one of ζ_3, ζ_3^2 precisely when $v_3(\mathrm{ord}_p\, 53) = 1$; it then gives the unique fixed degree-one branch. For an inert prime, the residue of 53 has prime-to-three order while the factor ζ_3 has order three, so neither component has a fixed branch. This proves (E123.6) and the stated unique branch. Norms of its prime ideals have residue weight one, so their classes agree under the actual maps m_d.

For clarity, the completed principal-divisor exactness is also literal. At a fixed field level let P be the group of principal outside-S divisors. It is a subgroup of a free abelian divisor group and is free. Thus $0 \to V_d \to L_d^\times \to P \to 0$ splits as abelian groups. The quotient of the divisor group by P is the finite S-class group. Its three-primary exponent is nine for $d \geq 2$. If $x \in P \cap 3^n \operatorname{Div}_{S^c}$, write $x = 3^n y$. Then $[y]$ is three-primary and killed by nine, so $9y \in P$ and $x \in 3^{n-2}P$ for $n \geq 2$. These two cofinal filtrations show that the completed image of P is exactly the kernel of the finite map to C_d. The completed divisor group is the restricted sequence group, not the full product with its product topology.

For $v \in \mathcal{R}(\mathcal{P})$, the resulting divisor at every level has class $c(v)$. Hence $c(v) = 0$ is necessary. It is sufficient as well. At each level the fibre of completed elements with that divisor is a nonempty compact torsor under $\widehat{V}_{d,3}$. Any finite chain of norm equations is solved by choosing an element at the highest level and taking its actual norms down the chain. The finite intersection property in the product of these compact fibres gives a compatible element at

every level. This proves (E123.10); it does not assume that unit norms are onto. The explicit prime basis proves the surjectivity of c and both inverse formulas in (E123.9).

Choose coherent lifts of the divisor basis as in (E123.11). Modulo 3^n at any fixed field level, all but finitely many exponents t_p vanish, so the infinite product converges. This also proves continuity, and the displayed inverse is continuous by division and the divisor map. If chosen lifts are replaced by $A_i y_i$, $y_i \in \mathcal{V}$, then $s'(v) = s(v)\prod_i y_i^{a_i(v)}$. The corresponding coordinate change is $(y, v) \mapsto (y\prod_i y_i^{-a_i(v)}, v)$. This identifies every coordinate change.

The compact metrizable group $\mathcal{V}$ is torsion-free. A torsion coherent family has every component in the literal μ_3, and every further vertical norm is cubing, so every fixed component is one. Its Pontryagin dual is therefore a countable divisible three-primary group. Choose a basis $\chi_{i,1}$ of its three-torsion vector space, and recursively choose $3\chi_{i,n+1} = \chi_{i,n}$. The map to a product of copies of $\mathbb{Z}_3$ is

$$\Phi(y)_i \bmod 3^n = 3^n \chi_{i,n}(y) \quad \text{in } \mathbb{Z}/3^n. \tag{E123.25}$$

The characters on the right have values in $3^{-n}\mathbb{Z}/\mathbb{Z}$. Their independence and generation follow by successively multiplying a finite-order character by its highest possible power of three and subtracting its basis lift. Thus the inverse is the unique point satisfying all these finite character equations; compactness gives existence. This proves the topological inverse, not merely an equality of ranks.

There are countably infinitely many factors. Apply Kato Theorem1.1(2) to $F = L_d$, $p = 3$, and $a_1 = 53$: the field has one place over three, and the actual extension $L_d(\mu_{3^\infty}, 53^{1/3^\infty})$ contains all later L_e. The theorem uses the coherent norm image as defined on its first page and proves finite index in the completed group of three-units. Restrict that actual coherent system to the fields L_e. Its image in V_d has rank 3^d. These unbounded ranks exclude a finite number of factors in (E123.25) [2]. Only this proved finite-index assertion is used; no index or unproved numerical unit basis is assigned.

Finally, the rational norm of a prime ideal in the unique surviving branch is p, and the two S-prime residue degrees are one and two. This proves every exponent in (E123.12). The family β_d is norm-compatible because the constant term of its odd-degree radical equation gives $N_{L_{d+1}/L_d}\beta_{d+1} = \beta_d$. For a first coordinate $3^k\kappa_{53}$, the positional factor under discussion is the actual $j = 1$ factor in E.46/E.111, with root coordinate sent to β_d along its cofinal subsystem; root cubing is its field norm. Thus, if $\mathcal{X}_{\neq 1}$ denotes the other actual factors, the full H-fibre with this factor expanded is exactly

$$\left\{(z, y, v) \in \mathcal{X}_{\neq 1} \times \mathcal{V} \times \mathcal{D} : N(z)\, 3^{\nu_3(y)+\rho(v)} 53^{\nu_{53}(y)+\eta(v)} \prod_p p^{v_p} \in (\widehat{\mathbb{Q}^\times}_3)^3\right\}. \tag{E123.26}$$

This is an equality through E.46's cocycle–norm maps, with their inverse, and retains the actual norms of the other factors. Substitution $z = 1$, $y = 1$, $v \mapsto 3v$ gives (E123.13). Its inverse is the outside-S divisor divided by three. In the product coordinates (E123.11) its image has unit coordinate one and divisor in the closed subgroup $3\mathcal{D}$, which proves closedness. The subgroup $3\mathcal{D}$ need not be replaced by $\mathcal{D} \cap 3\mathcal{R}(\mathcal{P})$; the displayed map uses precisely the former. The computation evaluates this positional factor, its coherent divisor image and the actual section subfamily. The remaining valuation functionals and other positional factors in (E123.26) are explicitly retained as maps, and no additional intrinsic lift is inferred.

The manuscript anchor is the same Fermat-tower and arithmetic splitting discussion, §51.16, folios733v–736v, scans172–178, used in E.23 and E.46. The code records all stopped attempts separately, including the upper flag-one certification timeout; the final ideal certification above uses the independent actual basis (E123.8) and literal ideal equalities. □

References

[1] J. S. Milne, *Class Field Theory*, version4.03, 2020.

[2] K. Kato, *Universal Norms of p-Units in Some Non-Commutative Galois Extensions*, Documenta Mathematica, Extra Volume Coates (2006), 551–565.